

SSS 1-3

INGREDIENT LAB ACTIVITY SYSTEM

IL-401 to IL-500

DESIGN -> INSTRUMENT -> PROCESS-MAP -> VALIDATE -> OPTIMIZE

Purpose	Build food and ingredient literacy by connecting plant structure, observation, measurement, post-harvest change, quality and product-development thinking.
Ingredient set	Hibiscus, ginger, turmeric, pineapple, citrus, mint, moringa, watermelon, tomato and maize.
Core rule	The Ingredient Lab is an educational investigation environment. It does not require tasting, medical claims or uncontrolled food processing.
Connections	Link each activity to GrowMeal vocabulary, journals, practical cards, Farm-to-Food activities, missions and assessment.

INGREDIENT LAB - HIBISCUS | PLANT PART: CALYX

Ten investigations develop progressively from ingredient identity and structure to quality, post-harvest change and an independent investigation challenge.

IL-401 - HIBISCUS - IDENTITY & PLANT PART

Plant part	Calyx
Learning objective	Design a research or product-development investigation of identity & plant part in hibiscus, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-402 - HIBISCUS - EXTERNAL STRUCTURE

Plant part	Calyx
Learning objective	Design a research or product-development investigation of external structure in hibiscus, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-403 - HIBISCUS - INTERNAL STRUCTURE

Plant part	Calyx
Learning objective	Design a research or product-development investigation of internal structure in hibiscus, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-404 - HIBISCUS - COLOUR & TEXTURE

Plant part	Calyx
Learning objective	Design a research or product-development investigation of colour & texture in hibiscus, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-405 - HIBISCUS - MASS & SIZE

Plant part	Calyx
Learning objective	Design a research or product-development investigation of mass & size in hibiscus, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-406 - HIBISCUS - WATER/CHANGE OBSERVATION

Plant part	Calyx
Learning objective	Design a research or product-development investigation of water/change observation in hibiscus, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-407 - HIBISCUS - SORTING & QUALITY

Plant part	Calyx
Learning objective	Design a research or product-development investigation of sorting & quality in hibiscus, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-408 - HIBISCUS - POST-HARVEST CHANGE

Plant part	Calyx
Learning objective	Design a research or product-development investigation of post-harvest change in hibiscus, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-409 - HIBISCUS - INGREDIENT-TO-FOOD PATHWAY

Plant part	Calyx
Learning objective	Design a research or product-development investigation of ingredient-to-food pathway in hibiscus, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-410 - HIBISCUS - INVESTIGATION CHALLENGE

Plant part	Calyx
Learning objective	Design a research or product-development investigation of investigation challenge in hibiscus, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

INGREDIENT LAB - GINGER | PLANT PART: RHIZOME

Ten investigations develop progressively from ingredient identity and structure to quality, post-harvest change and an independent investigation challenge.

IL-411 - GINGER - IDENTITY & PLANT PART

Plant part	Rhizome
Learning objective	Design a research or product-development investigation of identity & plant part in ginger, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
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Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-412 - GINGER - EXTERNAL STRUCTURE

Plant part	Rhizome
Learning objective	Design a research or product-development investigation of external structure in ginger, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
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AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-413 - GINGER - INTERNAL STRUCTURE

Plant part	Rhizome
Learning objective	Design a research or product-development investigation of internal structure in ginger, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
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Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-414 - GINGER - COLOUR & TEXTURE

Plant part	Rhizome
Learning objective	Design a research or product-development investigation of colour & texture in ginger, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-415 - GINGER - MASS & SIZE

Plant part	Rhizome
Learning objective	Design a research or product-development investigation of mass & size in ginger, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
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Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-416 - GINGER - WATER/CHANGE OBSERVATION

Plant part	Rhizome
Learning objective	Design a research or product-development investigation of water/change observation in ginger, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
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Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-417 - GINGER - SORTING & QUALITY

Plant part	Rhizome
Learning objective	Design a research or product-development investigation of sorting & quality in ginger, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
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Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-418 - GINGER - POST-HARVEST CHANGE

Plant part	Rhizome
Learning objective	Design a research or product-development investigation of post-harvest change in ginger, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-419 - GINGER - INGREDIENT-TO-FOOD PATHWAY

Plant part	Rhizome
Learning objective	Design a research or product-development investigation of ingredient-to-food pathway in ginger, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-420 - GINGER - INVESTIGATION CHALLENGE

Plant part	Rhizome
Learning objective	Design a research or product-development investigation of investigation challenge in ginger, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

INGREDIENT LAB - TURMERIC | PLANT PART: RHIZOME

Ten investigations develop progressively from ingredient identity and structure to quality, post-harvest change and an independent investigation challenge.

IL-421 - TURMERIC - IDENTITY & PLANT PART

Plant part	Rhizome
Learning objective	Design a research or product-development investigation of identity & plant part in turmeric, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-422 - TURMERIC - EXTERNAL STRUCTURE

Plant part	Rhizome
Learning objective	Design a research or product-development investigation of external structure in turmeric, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-423 - TURMERIC - INTERNAL STRUCTURE

Plant part	Rhizome
Learning objective	Design a research or product-development investigation of internal structure in turmeric, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-424 - TURMERIC - COLOUR & TEXTURE

Plant part	Rhizome
Learning objective	Design a research or product-development investigation of colour & texture in turmeric, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-425 - TURMERIC - MASS & SIZE

Plant part	Rhizome
Learning objective	Design a research or product-development investigation of mass & size in turmeric, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-426 - TURMERIC - WATER/CHANGE OBSERVATION

Plant part	Rhizome
Learning objective	Design a research or product-development investigation of water/change observation in turmeric, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-427 - TURMERIC - SORTING & QUALITY

Plant part	Rhizome
Learning objective	Design a research or product-development investigation of sorting & quality in turmeric, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-428 - TURMERIC - POST-HARVEST CHANGE

Plant part	Rhizome
Learning objective	Design a research or product-development investigation of post-harvest change in turmeric, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-429 - TURMERIC - INGREDIENT-TO-FOOD PATHWAY

Plant part	Rhizome
Learning objective	Design a research or product-development investigation of ingredient-to-food pathway in turmeric, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-430 - TURMERIC - INVESTIGATION CHALLENGE

Plant part	Rhizome
Learning objective	Design a research or product-development investigation of investigation challenge in turmeric, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

INGREDIENT LAB - PINEAPPLE | PLANT PART: FRUIT

Ten investigations develop progressively from ingredient identity and structure to quality, post-harvest change and an independent investigation challenge.

IL-431 - PINEAPPLE - IDENTITY & PLANT PART

Plant part	Fruit
Learning objective	Design a research or product-development investigation of identity & plant part in pineapple, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-432 - PINEAPPLE - EXTERNAL STRUCTURE

Plant part	Fruit
Learning objective	Design a research or product-development investigation of external structure in pineapple, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-433 - PINEAPPLE - INTERNAL STRUCTURE

Plant part	Fruit
Learning objective	Design a research or product-development investigation of internal structure in pineapple, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-434 - PINEAPPLE - COLOUR & TEXTURE

Plant part	Fruit
Learning objective	Design a research or product-development investigation of colour & texture in pineapple, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-435 - PINEAPPLE - MASS & SIZE

Plant part	Fruit
Learning objective	Design a research or product-development investigation of mass & size in pineapple, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-436 - PINEAPPLE - WATER/CHANGE OBSERVATION

Plant part	Fruit
Learning objective	Design a research or product-development investigation of water/change observation in pineapple, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-437 - PINEAPPLE - SORTING & QUALITY

Plant part	Fruit
Learning objective	Design a research or product-development investigation of sorting & quality in pineapple, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-438 - PINEAPPLE - POST-HARVEST CHANGE

Plant part	Fruit
Learning objective	Design a research or product-development investigation of post-harvest change in pineapple, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-439 - PINEAPPLE - INGREDIENT-TO-FOOD PATHWAY

Plant part	Fruit
Learning objective	Design a research or product-development investigation of ingredient-to-food pathway in pineapple, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-440 - PINEAPPLE - INVESTIGATION CHALLENGE

Plant part	Fruit
Learning objective	Design a research or product-development investigation of investigation challenge in pineapple, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

INGREDIENT LAB - CITRUS | PLANT PART: FRUIT

Ten investigations develop progressively from ingredient identity and structure to quality, post-harvest change and an independent investigation challenge.

IL-441 - CITRUS - IDENTITY & PLANT PART

Plant part	Fruit
Learning objective	Design a research or product-development investigation of identity & plant part in citrus, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-442 - CITRUS - EXTERNAL STRUCTURE

Plant part	Fruit
Learning objective	Design a research or product-development investigation of external structure in citrus, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-443 - CITRUS - INTERNAL STRUCTURE

Plant part	Fruit
Learning objective	Design a research or product-development investigation of internal structure in citrus, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-444 - CITRUS - COLOUR & TEXTURE

Plant part	Fruit
Learning objective	Design a research or product-development investigation of colour & texture in citrus, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-445 - CITRUS - MASS & SIZE

Plant part	Fruit
Learning objective	Design a research or product-development investigation of mass & size in citrus, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-446 - CITRUS - WATER/CHANGE OBSERVATION

Plant part	Fruit
Learning objective	Design a research or product-development investigation of water/change observation in citrus, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-447 - CITRUS - SORTING & QUALITY

Plant part	Fruit
Learning objective	Design a research or product-development investigation of sorting & quality in citrus, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-448 - CITRUS - POST-HARVEST CHANGE

Plant part	Fruit
Learning objective	Design a research or product-development investigation of post-harvest change in citrus, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-449 - CITRUS - INGREDIENT-TO-FOOD PATHWAY

Plant part	Fruit
Learning objective	Design a research or product-development investigation of ingredient-to-food pathway in citrus, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-450 - CITRUS - INVESTIGATION CHALLENGE

Plant part	Fruit
Learning objective	Design a research or product-development investigation of investigation challenge in citrus, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

INGREDIENT LAB - MINT | PLANT PART: LEAF

Ten investigations develop progressively from ingredient identity and structure to quality, post-harvest change and an independent investigation challenge.

IL-451 - MINT - IDENTITY & PLANT PART

Plant part	Leaf
Learning objective	Design a research or product-development investigation of identity & plant part in mint, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-452 - MINT - EXTERNAL STRUCTURE

Plant part	Leaf
Learning objective	Design a research or product-development investigation of external structure in mint, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-453 - MINT - INTERNAL STRUCTURE

Plant part	Leaf
Learning objective	Design a research or product-development investigation of internal structure in mint, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-454 - MINT - COLOUR & TEXTURE

Plant part	Leaf
Learning objective	Design a research or product-development investigation of colour & texture in mint, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-455 - MINT - MASS & SIZE

Plant part	Leaf
Learning objective	Design a research or product-development investigation of mass & size in mint, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-456 - MINT - WATER/CHANGE OBSERVATION

Plant part	Leaf
Learning objective	Design a research or product-development investigation of water/change observation in mint, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-457 - MINT - SORTING & QUALITY

Plant part	Leaf
Learning objective	Design a research or product-development investigation of sorting & quality in mint, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-458 - MINT - POST-HARVEST CHANGE

Plant part	Leaf
Learning objective	Design a research or product-development investigation of post-harvest change in mint, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-459 - MINT - INGREDIENT-TO-FOOD PATHWAY

Plant part	Leaf
Learning objective	Design a research or product-development investigation of ingredient-to-food pathway in mint, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-460 - MINT - INVESTIGATION CHALLENGE

Plant part	Leaf
Learning objective	Design a research or product-development investigation of investigation challenge in mint, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

INGREDIENT LAB - MORINGA | PLANT PART: LEAF

Ten investigations develop progressively from ingredient identity and structure to quality, post-harvest change and an independent investigation challenge.

IL-461 - MORINGA - IDENTITY & PLANT PART

Plant part	Leaf
Learning objective	Design a research or product-development investigation of identity & plant part in moringa, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-462 - MORINGA - EXTERNAL STRUCTURE

Plant part	Leaf
Learning objective	Design a research or product-development investigation of external structure in moringa, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-463 - MORINGA - INTERNAL STRUCTURE

Plant part	Leaf
Learning objective	Design a research or product-development investigation of internal structure in moringa, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-464 - MORINGA - COLOUR & TEXTURE

Plant part	Leaf
Learning objective	Design a research or product-development investigation of colour & texture in moringa, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-465 - MORINGA - MASS & SIZE

Plant part	Leaf
Learning objective	Design a research or product-development investigation of mass & size in moringa, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-466 - MORINGA - WATER/CHANGE OBSERVATION

Plant part	Leaf
Learning objective	Design a research or product-development investigation of water/change observation in moringa, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-467 - MORINGA - SORTING & QUALITY

Plant part	Leaf
Learning objective	Design a research or product-development investigation of sorting & quality in moringa, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-468 - MORINGA - POST-HARVEST CHANGE

Plant part	Leaf
Learning objective	Design a research or product-development investigation of post-harvest change in moringa, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-469 - MORINGA - INGREDIENT-TO-FOOD PATHWAY

Plant part	Leaf
Learning objective	Design a research or product-development investigation of ingredient-to-food pathway in moringa, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-470 - MORINGA - INVESTIGATION CHALLENGE

Plant part	Leaf
Learning objective	Design a research or product-development investigation of investigation challenge in moringa, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

INGREDIENT LAB - WATERMELON | PLANT PART: FRUIT

Ten investigations develop progressively from ingredient identity and structure to quality, post-harvest change and an independent investigation challenge.

IL-471 - WATERMELON - IDENTITY & PLANT PART

Plant part	Fruit
Learning objective	Design a research or product-development investigation of identity & plant part in watermelon, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-472 - WATERMELON - EXTERNAL STRUCTURE

Plant part	Fruit
Learning objective	Design a research or product-development investigation of external structure in watermelon, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-473 - WATERMELON - INTERNAL STRUCTURE

Plant part	Fruit
Learning objective	Design a research or product-development investigation of internal structure in watermelon, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-474 - WATERMELON - COLOUR & TEXTURE

Plant part	Fruit
Learning objective	Design a research or product-development investigation of colour & texture in watermelon, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-475 - WATERMELON - MASS & SIZE

Plant part	Fruit
Learning objective	Design a research or product-development investigation of mass & size in watermelon, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-476 - WATERMELON - WATER/CHANGE OBSERVATION

Plant part	Fruit
Learning objective	Design a research or product-development investigation of water/change observation in watermelon, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-477 - WATERMELON - SORTING & QUALITY

Plant part	Fruit
Learning objective	Design a research or product-development investigation of sorting & quality in watermelon, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-478 - WATERMELON - POST-HARVEST CHANGE

Plant part	Fruit
Learning objective	Design a research or product-development investigation of post-harvest change in watermelon, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-479 - WATERMELON - INGREDIENT-TO-FOOD PATHWAY

Plant part	Fruit
Learning objective	Design a research or product-development investigation of ingredient-to-food pathway in watermelon, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-480 - WATERMELON - INVESTIGATION CHALLENGE

Plant part	Fruit
Learning objective	Design a research or product-development investigation of investigation challenge in watermelon, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

INGREDIENT LAB - TOMATO | PLANT PART: FRUIT

Ten investigations develop progressively from ingredient identity and structure to quality, post-harvest change and an independent investigation challenge.

IL-481 - TOMATO - IDENTITY & PLANT PART

Plant part	Fruit
Learning objective	Design a research or product-development investigation of identity & plant part in tomato, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-482 - TOMATO - EXTERNAL STRUCTURE

Plant part	Fruit
Learning objective	Design a research or product-development investigation of external structure in tomato, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-483 - TOMATO - INTERNAL STRUCTURE

Plant part	Fruit
Learning objective	Design a research or product-development investigation of internal structure in tomato, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-484 - TOMATO - COLOUR & TEXTURE

Plant part	Fruit
Learning objective	Design a research or product-development investigation of colour & texture in tomato, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-485 - TOMATO - MASS & SIZE

Plant part	Fruit
Learning objective	Design a research or product-development investigation of mass & size in tomato, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-486 - TOMATO - WATER/CHANGE OBSERVATION

Plant part	Fruit
Learning objective	Design a research or product-development investigation of water/change observation in tomato, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-487 - TOMATO - SORTING & QUALITY

Plant part	Fruit
Learning objective	Design a research or product-development investigation of sorting & quality in tomato, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-488 - TOMATO - POST-HARVEST CHANGE

Plant part	Fruit
Learning objective	Design a research or product-development investigation of post-harvest change in tomato, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-489 - TOMATO - INGREDIENT-TO-FOOD PATHWAY

Plant part	Fruit
Learning objective	Design a research or product-development investigation of ingredient-to-food pathway in tomato, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-490 - TOMATO - INVESTIGATION CHALLENGE

Plant part	Fruit
Learning objective	Design a research or product-development investigation of investigation challenge in tomato, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

INGREDIENT LAB - MAIZE | PLANT PART: SEED/GRAIN

Ten investigations develop progressively from ingredient identity and structure to quality, post-harvest change and an independent investigation challenge.

IL-491 - MAIZE - IDENTITY & PLANT PART

Plant part	Seed/Grain
Learning objective	Design a research or product-development investigation of identity & plant part in maize, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-492 - MAIZE - EXTERNAL STRUCTURE

Plant part	Seed/Grain
Learning objective	Design a research or product-development investigation of external structure in maize, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-493 - MAIZE - INTERNAL STRUCTURE

Plant part	Seed/Grain
Learning objective	Design a research or product-development investigation of internal structure in maize, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-494 - MAIZE - COLOUR & TEXTURE

Plant part	Seed/Grain
Learning objective	Design a research or product-development investigation of colour & texture in maize, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-495 - MAIZE - MASS & SIZE

Plant part	Seed/Grain
Learning objective	Design a research or product-development investigation of mass & size in maize, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-496 - MAIZE - WATER/CHANGE OBSERVATION

Plant part	Seed/Grain
Learning objective	Design a research or product-development investigation of water/change observation in maize, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-497 - MAIZE - SORTING & QUALITY

Plant part	Seed/Grain
Learning objective	Design a research or product-development investigation of sorting & quality in maize, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-498 - MAIZE - POST-HARVEST CHANGE

Plant part	Seed/Grain
Learning objective	Design a research or product-development investigation of post-harvest change in maize, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-499 - MAIZE - INGREDIENT-TO-FOOD PATHWAY

Plant part	Seed/Grain
Learning objective	Design a research or product-development investigation of ingredient-to-food pathway in maize, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-500 - MAIZE - INVESTIGATION CHALLENGE

Plant part	Seed/Grain
Learning objective	Design a research or product-development investigation of investigation challenge in maize, integrating measurement quality, process flow, validation and optimization.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define performance objective and acceptance criteria; specify sample/process boundary; establish measurement/QC protocol; collect and model data; validate against reference evidence; compare scenarios; recommend optimization.
Evidence to record	Protocol, traceable dataset, QC/validation record, model or quantitative comparison, uncertainty and optimization recommendation.
Questions	Which decision must the evidence support? How is measurement quality verified? What trade-off constrains the preferred process or specification?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?